

Day 1 Practice Sheet — Excel Formulas & IF Function

Module: Excel for Data Analysis — Formulas & IF

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Objective: Practise SUM, AVERAGE, COUNT, COUNTA, MIN, MAX, IF and nested IF on a sales dataset.

SAMPLE DATASET

Use the sample data table below for all questions. The table shows Order ID, Salesperson, City, Product, Quantity, Unit Price, and Total Sales.

Order ID	Salesperson	City	Product	Quantity	Unit Price	Total Sales
ORD-001	Amit Sharma	Delhi	Laptop	2	45000	
ORD-002	Priya Singh	Mumbai	Mobile	5	18000	
ORD-003	Rahul Verma	Chennai	Tablet	3	22000	
ORD-004	Neha Gupta	Pune	Laptop	1	45000	
ORD-005	Amit Sharma	Delhi	Mobile	4	18000	
ORD-006	Priya Singh	Mumbai	Laptop	2	45000	
ORD-007	Rohit Jain	Jaipur	Tablet	6	22000	
ORD-008	Neha Gupta	Pune	Mobile	3	18000	
ORD-009	Rahul Verma	Chennai	Laptop	1	45000	
ORD-010	Rohit Jain	Jaipur	Mobile	2	18000	

Q1 [SUM]

In cell G2, write a formula to calculate Total Sales for ORD-001.

Copy it down to G11. Total Sales = Quantity × Unit Price.

Total Sales
90000
90000
66000
45000
72000
90000
132000
54000
45000
36000

Q2 [SUM]

In a new cell below the table, calculate the Grand Total of all sales using SUM.

Grand Total	720000
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Q3 [AVERAGE]

Calculate the Average Order Value across all 10 orders using AVERAGE on the Total Sales column.

Average order value	72000
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Q4 [COUNT/COUNTA]

Use COUNT on the Quantity column. Then use COUNTA on the Salesperson column. Are the results the same? Why?

count on Quantity	10
counta on salesperson	10

Q5 [MIN/MAX]

Find the Minimum and Maximum values in the Unit Price column.

Min	Max
18000	45000

Q6
[MIN/MAX]

Which order had the highest Total Sales? Write MAX and note the Order ID manually.

Order ID	highest total sales
ORD-007	90000

Q7 [IF]

In a new column called 'Performance', write an IF formula: if Total Sales > 80000 then 'High', else 'Low'.

Order ID	Performance
ORD-001	High
ORD-002	High
ORD-003	Low
ORD-004	Low
ORD-005	Low
ORD-006	High
ORD-007	High
ORD-008	Low
ORD-009	Low
ORD-010	Low

Q8 [IF]

In a new column called 'City Category', write IF: if City = 'Delhi' or 'Mumbai' then 'Metro' else 'Non-Metro'. Hint: use IF with OR().

Order ID	City	City Category
ORD-001	Delhi	Metro
ORD-002	Mumbai	Metro
ORD-003	Chennai	Non-Metro
ORD-004	Pune	Non-Metro
ORD-005	Delhi	Metro
ORD-006	Mumbai	Metro
ORD-007	Jaipur	Non-Metro
ORD-008	Pune	Non-Metro
ORD-009	Chennai	Non-Metro
ORD-010	Jaipur	Non-Metro

Q9 [Nested IF]

In a new column called 'Commission Rate', write a nested IF: Total Sales > 100000 → 10%; Total Sales > 50000 → 7%; else 5%.

Order ID	Order ID	Commision Rate
ORD-001	ORD-001	7%
ORD-002	ORD-002	7%
ORD-003	ORD-003	7%
ORD-004	ORD-004	5%
ORD-005	ORD-005	7%
ORD-006	ORD-006	7%
ORD-007	ORD-007	10%
ORD-008	ORD-008	7%
ORD-009	ORD-009	5%
ORD-010	ORD-010	5%

Q10 [Nested IF]

Add a column 'Commission Amount' = Total Sales × Commission Rate. Then use SUM to find total commission payout.

Order ID	Order ID	Commision Rate	Commision Amount
ORD-001	ORD-001	7%	6300
ORD-002	ORD-002	7%	6300
ORD-003	ORD-003	7%	4620
ORD-004	ORD-004	5%	2250
ORD-005	ORD-005	7%	5040
ORD-006	ORD-006	7%	6300
ORD-007	ORD-007	10%	13200
ORD-008	ORD-008	7%	3780
ORD-009	ORD-009	5%	2250
ORD-010	ORD-010	5%	1800

BONUS

Highlight all cells in 'Performance' column where value is 'High' using Conditional Formatting (Home → Conditional Formatting → Highlight Cells)

Order ID	Salesperson	City	Product	Quantity	Unit Price	Order ID	Performance	City Category	Commision Rate	Commision Amount
ORD-001	Amit Sharma	Delhi	Laptop	2	45000	ORD-001	High	Metro	7%	6300
ORD-002	Priya Singh	Mumbai	Mobile	5	18000	ORD-002	High	Metro	7%	6300
ORD-003	Rahul Verma	Chennai	Tablet	3	22000	ORD-003	Low	Non-Metro	7%	4620
ORD-004	Neha Gupta	Pune	Laptop	1	45000	ORD-004	Low	Non-Metro	5%	2250
ORD-005	Amit Sharma	Delhi	Mobile	4	18000	ORD-005	Low	Metro	7%	5040
ORD-006	Priya Singh	Mumbai	Laptop	2	45000	ORD-006	High	Metro	7%	6300
ORD-007	Rohit Jain	Jaipur	Tablet	6	22000	ORD-007	High	Non-Metro	10%	13200
ORD-008	Neha Gupta	Pune	Mobile	3	18000	ORD-008	Low	Non-Metro	7%	3780
ORD-009	Rahul Verma	Chennai	Laptop	1	45000	ORD-009	Low	Non-Metro	5%	2250
ORD-010	Rohit Jain	Jaipur	Mobile	2	18000	ORD-010	Low	Non-Metro	5%	1800
					Grand Total	720000			Total Commision Payout	51840